No.	Hands-on Assignment	Topics Covered	Status
1	Write a Program that accepts two Strings as command line arguments and generate the output in the required format. Example1) If the two command line arguments are Wipro and Bangalore then the output generated should be Wipro Technologies Bangalore. Example2) If the command line arguments are ABC and Mumbai then the output generated should be ABC Technologies Mumbai [Note: It is mandatory to pass two arguments in command line]	Command Line Argument	☐
2	Write a Program to accept a String as a command line argument and print a Welcome message as given below. Example1) C:\> java Sample John O/P Expected : Welcome John	Command Line Argument	☐
3	Write a Program to accept two integers as command line arguments and print the sum of the two numbers Example1) C:\>java Sample 10 20 O/P Expected : The sum of 10 and 20 is 30	Command Line Argument	☐

Hands-on Assignments for Classes and Objects

Complete the below hands-on assignments before proceeding with the next Topic

No.	Hands-on Assignment	Topics Covered	Status
1	Create a class Box that uses a parameterized constructor to initialize the dimensions of a box.The dimensions of the Box are width, height, depth. The class should have a method that can return the volume of the box. Create an object of the Box class and test the functionalities.	Classes and Objects, Constructor	☐
2	Create a new class called Calculator with the following methods: 1. A static method called powerInt(int num1,int num2) This method should return num1 to the power num2. 2. A static method called powerDouble(double num1,int num2). This method should return num1 to the power num2. 3. Invoke both the methods and test the functionalities. Hint: Use Math.pow(double,double) to calculate the power.	Classes and Objects, Constructor, static	☐
3	Design a class that can be used by a health care professional to keep track of a patient's vital statistics. The following are the details. Name of the class - Patient Member Variables - patientName(String),height(double),width(double) Member Function - double computeBMI() The above method should compute the BMI and return the result. The formula for computation of BMI is weight (in kg) ÷ height*height(in metres). Create an object of the Patient class and check the results.	Classes and Objects, Constructor, static	☐

Hands-on Assignments for Encapsulation / Abstraction

Complete the below hands-on assignments before proceeding with the next Topic

No.	Hands-on Assignment	Topics Covered	Status
1	Create a class Author with the following information. Member variables : name (String), email (String), and gender (char) Parameterized Constructor: To initialize the variables Create a class Book with the following information. Member variables : name (String), author (of the class Author you have just created), price (double), and qtyInStock (int) [Assumption: Each book will be written by exactly one Author] Parameterized Constructor: To initialize the variables Getters and Setters for all the member variables In the main method, create a book object and print all details of the book (including the author details)	Encapsulation / Abstraction	

Hands-on Assignments for Inheritance

Complete the below hands-on assignments before proceeding with the next Topic

No.	Hands-on Assignment	Topics Covered	Status
1	Create a class named 'Animal' which includes methods like eat() and sleep(). Create a child class of Animal named 'Bird' and override the parent class methods. Add a new method named fly(). Create an instance of Animal class and invoke the eat and sleep methods using this object. Create an instance of Bird class and invoke the eat, sleep and fly methods using this object.	Inheritance	☐
2	Create a class called Person with a member variable name. Save it in a file called Person.java Create a class called Employee that will inherit the Person class.The other data members of the Employee class are annual salary (double), the year the employee started to work, and the national insurance number which is a String.Save this in a file called Employee.java Your class should have the necessary constructors and getter/setter methods. Write another class called TestEmployee, containing a main method to fully test your class definition.	Inheritance	☐
3	Create a school application with a class called Person. Create name and dateOfBirth as member variables. Create a class called Teacher that inherits from the Person class. The teacher will have additional properties like salary, and the subject that the teacher teaches. Create a class called Student that inherits from Person class. This class will have a member variable called studentId. Create a class called College Student that inherits from Student class. This class will have collegeName, the year in which the student is studying (first/second/third/fourth) etc. Create objects of each of this classes, invoke and test the methods that are available in these classes.	Inheritance	☐

Hands-on Assignments for Overriding / Polymorphism

Complete the below hands-on assignments before proceeding with the next Topic

No.	Hands-on Assignment	Topics Covered	Status
1	Create a base class Fruit with name ,taste and size as its attributes. Create a method called eat() which describes the name of the fruit and its taste. Inherit the same in 2 other classes Apple and Orange and override the eat() method to represent each fruit taste.	Inheritance / Overriding	☐
2	Write a program to create a class named shape. It should contain 2 methods, draw() and erase() that prints “Drawing Shape” and “Erasing Shape” respectively. For this class, create three sub classes, Circle, Triangle and Square and each class should override the parent class functions - draw () and erase (). The draw() method should print “Drawing Circle”, “Drawing Triangle” and “Drawing Square” respectively. The erase() method should print “Erasing Circle”, “Erasing Triangle” and “Erasing Square” respectively. Create objects of Circle, Triangle and Square in the following way and observe the polymorphic nature of the class by calling draw() and erase() method using each object. Shape c=new Circle(); Shape t=new Triangle(); Shape s=new Square();	Polymorphism	☐

Hands-on Assignments for Interfaces

Complete the below hands-on assignments before proceeding with the next Topic

No.	Hands-on Assignment	Topics Covered	Status
1	An online library application need to be created for two types of users/roles-Adults and children. Both of these users should be able to register an account. Any user who is less than 12 years of age will be registered as a child and they can borrow a “Kids” category book for 10 days, whereas an adult can borrow “Fiction” category books which need to be returned within 7 days. Note: In future, more users/roles might be added to the library where similar rules will be enforced. Develop Interfaces and classes for the categories mentioned above. 1. Create an interface LibraryUser with the following methods void registerAccount() void requestBook () 2. Create 2 classes “KidUsers” and “AdultUser” which implements the LibraryUser interface. 3. Both the classes should have two instance variables - age(int),bookType(String) 4. The methods in the KidUser class should perform the following logic. registerAccount(): if age < 12, a message displaying “You have successfully registered under a Kids Account” should be displayed in the console. If(age>12), a message displaying, “Sorry, Age must be less than 12 to register as a kid” should be displayed in the console. requestBook():	Interfaces	☐

	if bookType is “Kids”, a message displaying “Book Issued successfully, please return the book within 10 days” should be displayed in the console,else, a message displaying, “Oops, you are allowed to take only kids books” should be displayed in the console. 5. The methods in the AdultUser class should perform the following logic. registerAccount(): if age > 12, a message displaying “You have successfully registered under an Adult Account” should be displayed in the console. If age<12, a message displaying, “Sorry, Age must be greater than 12 to register as an adult” should be displayed in the console. requestBook function: if bookType is “Fiction”, a message displaying “Book Issued successfully, please return the book within 7 days” should be displayed in the console., else, a message displaying, “Oops, you are allowed to take only adult Fiction books” should be displayed in the console. 6. Create a class “LibraryInterfaceDemo.java” with a main method and test the functionalities by creating objects of KidUser and AdultUser classes.		
2	Write an interface called Playable, with a method void play(); Let this interface be placed in a package called music. Write a class called Veena which implements Playable interface. Let this class be placed in a package music.string Write a class called Saxophone which implements Playable interface. Let this class be placed in a package music.wind Write another class Test in a package called live. Then, a. Create an instance of Veena and call play() method b. Create an instance of Saxophone and call play() method	Interfaces	☐

	c. Place the above instances in a variable of type Playable and then call play()		

Hands-on Assignments for Packages

Complete the below hands-on assignments before proceeding with the next Topic

No.	Hands-on Assignment	Topics Covered	Status
1	Create a package called test package. Define a class called foundation inside the test package. Inside the class, you need to define 4 integer variables: var1 with private access modifier var2 with default access modifier var3 with protected access modifier var4 with public access modifier Import this class and packages in another class. Try to access all 4 variables of the foundation class and see what variables are accessible and what are not accessible.	Packages Access control Using package	☐
2	Create a class called compartment which represents the ship compartments with attributes like height, width and breadth. Take care it should not conflict with the compartment class you have created in Abstract class exercise 2. To avoid conflict create this class in a new package called com.wipro.automobile.ship	Packages User defined packages	☐
3	Create a package called com.automobile. Define an abstract class called Vehicle. Vehicle class has the following abstract methods: public String getModelName() public String getRegistrationNumber() public String getOwnerName() Create twowheeler subpackage under automobile package Hero class extends automobile.vehicle class with the following methods public int getSpeed() - returns the current speed of the vehicle. public void radio() - provides facility to control the radio device Honda class extends com.automobile.vehicle class with the following methods public int getSpeed() - Returns the current speed of the vehicle. public void cdplayer() - provides facility to control the cd player device which is available in the car. Create a test class to test the methods available in all these child class.	Packages User defined packages	☐

4	Add the following ideas to the previous hands on: Create FourWheeler subpackage under automobile package Logan class extends com.automobile.Vehicle class public int speed() - Returns the current speed of the vehicle. public int gps() - provides facility to control the gps device Ford class extends com.automobile.Vehicle class public int speed() - Returns the current speed of the vehicle. public int tempControl() - provides facility to control the air conditioning device which is available in the car Create objects of the relevant classes and test the various functionalities of the class.	Packages User defined packages	☐